

Auditing Local LLM Workflows for Older-Adult Referral Support: Evidence Attribution, Model Sensitivity, Communication Stress Testing, and Data-Platform Implications

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Abstract

Large language models (LLMs) are increasingly explored for clinical decision support, but older-adult referral support remains a high-risk setting. Intake histories may be incomplete, caregiver-mediated, cognitively fragmented, or affected by underreported falls. This study audits local LLM workflows for older-adult intake-stage specialist referral prediction using MIMIC-IV-Ext-CDS, a MIMIC-derived benchmark for triage, referral, and diagnosis. We constructed an older-adult analytic cohort of 801 cases and compared majority and TF-IDF logistic-regression baselines with local Qwen2.5-1.5B workflows using raw intake text, standardized geriatric profiles, within-case evidence attribution, and abstention. We further performed local model-sensitivity analysis using Qwen2.5-3B and Phi-4-mini under raw and evidence-attributed prompting.

TF-IDF logistic regression achieved the strongest clean-condition performance, with accuracy 0.645, macro-F1 0.588, and top-3 recall 0.884. The strongest local LLM condition was Qwen2.5-3B raw prompting, with accuracy 0.537 and macro-F1 0.447. Phi-4-mini raw prompting achieved accuracy 0.496 and macro-F1 0.432. Within-case evidence attribution improved citation traceability but did not improve clean referral classification. Communication stress testing showed that caregiver-mediated histories produced the largest degradation, with profile macro-F1 decreasing by 0.090 and evidence-attributed macro-F1 decreasing by 0.080. These findings suggest that older-adult clinical LLM workflows should be evaluated against strong lexical baselines, tested across local model families, and stress-tested under communication degradation before deployment claims are made. The caregiver-mediated failure mode further motivates future integrated health-data platforms that reduce over-reliance on a single subjective intake narrative.

Keywords

clinical NLP, local LLM, MIMIC-IV-Ext-CDS, specialist referral, older adults, evidence attribution, model sensitivity, robustness, healthcare AI

1 Introduction

1.1 Motivation: Older-Adult Referral Support as High-Risk AI

Older-adult referral support is a high-risk setting for clinical AI. Older patients may present with multimorbidity, atypical symptoms, sensory barriers, cognitive impairment, incomplete recall, caregiver-mediated history, or underreported falls. These factors can degrade the subjective history available during early intake and

referral decisions. Prior work has documented that fall injuries are frequently underreported among older adults, with one study reporting that 72% of patients with Medicare-reimbursed fall-related injuries failed to report a fall injury when asked [6]. Communication challenges are also documented in emergency department encounters involving persons living with dementia and caregivers, where history-gathering and decision-making may depend on proxy communication and caregiver interpretation [2].

LLMs have demonstrated strong performance on medical question-answering benchmarks. Med-PaLM and related work showed that LLMs can encode clinical knowledge and perform well on medical question-answering datasets, while emphasizing that real clinical applications require more than benchmark accuracy [16, 17]. GPT-4 has also been evaluated on medical challenge problems, demonstrating strong capability but still requiring careful task-specific evaluation before clinical use [12]. These studies motivate clinical LLM exploration, but they do not eliminate the need for workflow-specific validation, baseline comparison, robustness testing, and transparent reporting.

This study examines specialist referral prediction as a controlled clinical NLP task for older-adult cases. It uses MIMIC-IV-Ext-CDS, a benchmark designed to evaluate LLM workflows for clinical decision support, including triage, referral, and diagnosis [4]. Related work by Gaber et al. evaluated LLM workflows for triage, referral, and diagnosis using MIMIC-derived cases [5]. Our study asks a narrower and complementary question: how do local LLM referral-support workflows behave in older-adult cases when patient communication is systematically degraded, and do larger or cross-family local models change the conclusion?

1.2 Study Objectives

The objective is not to build an autonomous referral agent. Instead, the objective is to audit local LLM referral-support workflows under a constrained, intake-stage, older-adult setting. Specifically, the project aims to:

- (1) construct an older-adult subset of MIMIC-IV-Ext-CDS for intake-stage specialist referral prediction;
- (2) compare local LLM prompting workflows against majority and TF-IDF logistic-regression baselines;
- (3) evaluate whether standardized geriatric profiles and within-case evidence attribution improve referral prediction;
- (4) test whether local model scaling or cross-family substitution changes the clean-condition findings;
- (5) evaluate deterministic older-adult communication perturbations; and

- (6) separate citation-ID validity from clinical evidence support.

1.3 Research Questions

The study is intentionally high-risk and audit-oriented. We do not assume that a local LLM will outperform classical text classification. We also do not assume that evidence-attributed prompting improves prediction. The research questions are:

RQ1. How do local LLM referral-support workflows compare with majority and TF-IDF logistic-regression baselines for older-adult intake-stage referral prediction?

RQ2. Do standardized geriatric profiles and within-case evidence-attributed prompts improve clean referral prediction compared with raw intake text?

RQ3. Does local model scaling or cross-family model substitution improve performance, as tested using Qwen2.5-1.5B, Qwen2.5-3B, and Phi-4-mini?

RQ4. Which older-adult communication perturbations most strongly degrade referral-support performance?

RQ5. Does within-case evidence attribution buffer the model against degraded subjective communication?

RQ6. What do the observed failure modes imply for future integrated health-data platforms for aging-society care?

1.4 Contributions

This study contributes:

- (1) a reproducible older-adult referral benchmark derived from MIMIC-IV-Ext-CDS;
- (2) a baseline-controlled comparison of local LLM prompting against TF-IDF logistic regression;
- (3) a standardized geriatric profile representation for intake-stage referral support;
- (4) a within-case evidence-attribution workflow distinct from external retrieval-augmented generation (RAG);
- (5) a local model-sensitivity analysis across Qwen2.5-1.5B, Qwen2.5-3B, and Phi-4-mini; and
- (6) a deterministic communication stress test motivated by older-adult care challenges.

The main contribution is an audit-style evaluation framework rather than a deployment-ready clinical tool.

2 Background and Motivation

2.1 MIMIC-IV and MIMIC-IV-Ext-CDS

MIMIC-IV is a large, publicly available electronic health record dataset sourced from Beth Israel Deaconess Medical Center [7]. MIMIC-IV-Ext-CDS extends MIMIC-derived data into a benchmark for LLM clinical decision support, including triage, specialist referral, and diagnosis. Its PhysioNet documentation reports 9,150 real-world medical cases and 2,200 refined task-specific cases [4]. Gaber et al. benchmarked LLM workflows across triage, referral, and diagnosis tasks using MIMIC-derived cases [5]. Our study differs by focusing on older-adult cases, local open-weight models, TF-IDF baseline comparison, and deterministic communication stress testing.

2.2 Medical LLM Evaluation Beyond Benchmark Accuracy

Medical LLM research has moved from multiple-choice medical question answering toward workflow evaluation. Med-PaLM introduced MultiMedQA and human evaluation axes such as factuality, reasoning, harm, and bias [16]. Later work improved medical question-answering performance but continued to emphasize that long-form answers and real-world workflows remain challenging [17]. This study follows that lesson by evaluating baselines, output-format reliability, task-specific metrics, robustness to input variation, and auditability.

2.3 Local LLM Deployment and Data Governance

For MIMIC-derived data, local deployment is central. PhysioNet guidance states that credentialed MIMIC data should not be shared with third parties through APIs or online platforms and strongly recommends local LLM deployment to maintain control over the data [13]. We therefore used local GPU inference rather than sending MIMIC-derived text to external LLM APIs.

The core local model was Qwen2.5-1.5B-Instruct. Model-sensitivity experiments included Qwen2.5-3B-Instruct and Phi-4-mini-instruct. Qwen2.5 is an open-weight instruction-tuned model family with multiple sizes and improved instruction-following capabilities [14, 15]. Phi-4-mini is a compact instruction model in the Phi-4 family, described as a lightweight open model designed for reasoning-dense data and memory-constrained settings [10, 11]. Our use of these models is a practical local-compute comparison, not a claim that these are clinically optimal models.

2.4 Evidence Attribution and RAG

RAG was proposed to combine parametric model memory with non-parametric retrieval for knowledge-intensive tasks [8]. Healthcare RAG studies often retrieve external guidelines, clinical documents, or prior cases to improve factual consistency and reduce hallucination [1]. Our method is narrower. It does not retrieve external guidelines, PubMed abstracts, or similar prior cases. It retrieves sections from the same patient case and asks the LLM to cite section IDs. We therefore describe the method as *within-case evidence attribution*, not external RAG. Citation-ID validity means that cited section IDs appeared in the retrieved evidence prompt; it does not prove that the evidence clinically supports the prediction.

2.5 Older-Adult Communication Stress

Older-adult communication can be clinically complex. Emergency department communication studies involving persons living with dementia and caregivers report challenges in history-taking and decision-making [2]. Fall reporting may also be unreliable among older adults [6]. These findings motivate deterministic stress-test perturbations: poor recall, mild confusion, caregiver-mediated reporting, and fall underreporting. We do not claim to simulate Japanese patients or validate real caregiver communication. Instead, we implement reproducible perturbations that operationalize plausible older-adult communication stressors.

2.6 Reporting Standards for Healthcare AI

Transparent reporting is essential for healthcare AI. TRIPOD+AI provides updated guidance for reporting prediction models that use regression or machine learning methods [3]. CONSORT-AI emphasizes clear reporting of AI interventions, inputs, outputs, and human-AI interaction in clinical-trial contexts [9]. Although this study is not a clinical trial and not a deployed intervention, these standards motivate explicit cohort definition, information-boundary control, baseline comparisons, error analysis, and limitation reporting.

3 Dataset and Study Context

This study uses MIMIC-IV-Ext-CDS version 1.0.2 as a controlled benchmark for specialist-referral prediction. The older-adult analytic cohort is restricted to cases with age ≥ 65 , valid clinical referral labels, and available chief complaint or HPI. The dataset is useful for reproducible audit-style evaluation, but it is not a Japanese clinical dataset. Therefore, Japan and other super-aging societies motivate the research context, while the empirical validation remains limited to this MIMIC-derived benchmark.

The study intentionally uses an intake-stage information boundary. Chief complaint, HPI, PMH when available, and vital signs are used for the main representation. Labs/tests are retained for availability analysis but excluded from the main intake-stage representation. This design supports a focused evaluation of early referral-support behavior and avoids making claims based on information that may not be available at the intended decision point.

4 Methods

4.1 Overall Workflow

The experiment follows an audit-oriented workflow. Official MIMIC-IV-Ext-CDS CSV files were merged into a case-level specialist-referral dataset. The analytic cohort was restricted to older-adult cases with valid clinical referral labels and available chief complaint or HPI. The label space was defined using the training split only by selecting the top eight referral labels and collapsing remaining labels into *other*. The same train, validation, and test splits were used across baselines and LLM arms. Each case was represented as raw intake text, a standardized geriatric profile, or a within-case evidence-attributed prompt. Majority and TF-IDF logistic-regression baselines were compared with local LLM prompting workflows. Finally, communication stress testing was applied to evaluate model behavior under degraded subjective history.

Figure 1 summarizes the core workflow. The figure shows the main baseline, representation, LLM, stress-test, and output layers. The additional model-sensitivity track is summarized separately in Table 3.

4.2 Target Decision Point and Input Boundary

The experiment uses an intake-stage information boundary. The main representation includes demographics, chief complaint, HPI, PMH when available, and vital signs. Labs/tests were retained for availability analysis but excluded from the main intake-stage representation. This design approximates an early referral-support

setting and reduces the risk of evaluating a model using information that may not be available at the intended decision point.

4.3 Data Source and Cohort Construction

We used MIMIC-IV-Ext-CDS version 1.0.2 as the benchmark source. The pipeline merged official CSV files covering specialty referral, initial assessment, clinical data, vital signs, patient demographics, triage level, and diagnosis. The base specialist-referral file was used rather than restricting to the optional clinician-approved subset. The merged specialist-referral subset contained 2,200 cases.

The final analytic cohort was restricted to cases with age ≥ 65 , non-empty referral labels, non-empty chief complaint or HPI, and valid clinical referral labels. Invalid non-clinical labels such as “no answer,” “unknown,” “none,” “not available,” “not applicable,” and missing values were removed. The final analytic cohort contained 801 cases. Cohort-flow reconstruction confirmed 2,200 initial rows, 817 cases with age ≥ 65 , and 801 cases after removal of 16 invalid clinical labels. The final cohort had 801 unique case IDs and 801 unique subject IDs, indicating no repeated-subject leakage in this cohort. The data were split into 560 training, 120 validation, and 121 test cases.

4.4 Label Engineering

Specialty referral labels may contain multiple specialties. For primary multiclass classification, we used the first listed specialty as the primary label. We also parsed the full normalized referral-label list for top-3 evaluation. The top eight referral labels were learned from the training split only, and labels outside this set were collapsed to *other*. This train-only label-space definition avoids using validation or test label distributions to define the output space.

Accuracy and macro-F1 on the primary collapsed label are the main performance metrics. LLM top-3 recall is reported as *post-processed exploratory top-3 recall*. In the LLM parser, the final parsed label is forced into the first ranked position to repair cases where the model output violated its own instruction that the first ranked label should equal the final label. Therefore, LLM top-3 recall should be interpreted as an exploratory ranking-support measure rather than a pure model-native ranking metric.

4.5 Case Representations

We evaluated three main case representations.

Raw intake text. The raw representation concatenates chief complaint, HPI, PMH when available, and vitals.

Standardized geriatric profile. The standardized profile restructures each case into explicit sections: demographics, chief complaint, HPI, PMH when available, vitals, and heuristic geriatric cues. These cues include fall, cognitive/confusion, mobility, and frailty signals. They are heuristic text-derived indicators, not validated frailty measurements. Because PMH and medication fields were unavailable in the merged representation, comorbidity-count and medication-count cues should be interpreted as missing-field artifacts rather than clinically reliable measurements.

Within-case evidence-attributed prompt. The evidence-attributed arm retrieves sections from the same patient case and instructs the LLM to cite section IDs. Retrieval used within-case sections only, including chief complaint, HPI, PMH when available,

End-to-End Experimental Architecture for Older-Adult Referral Support Evaluation

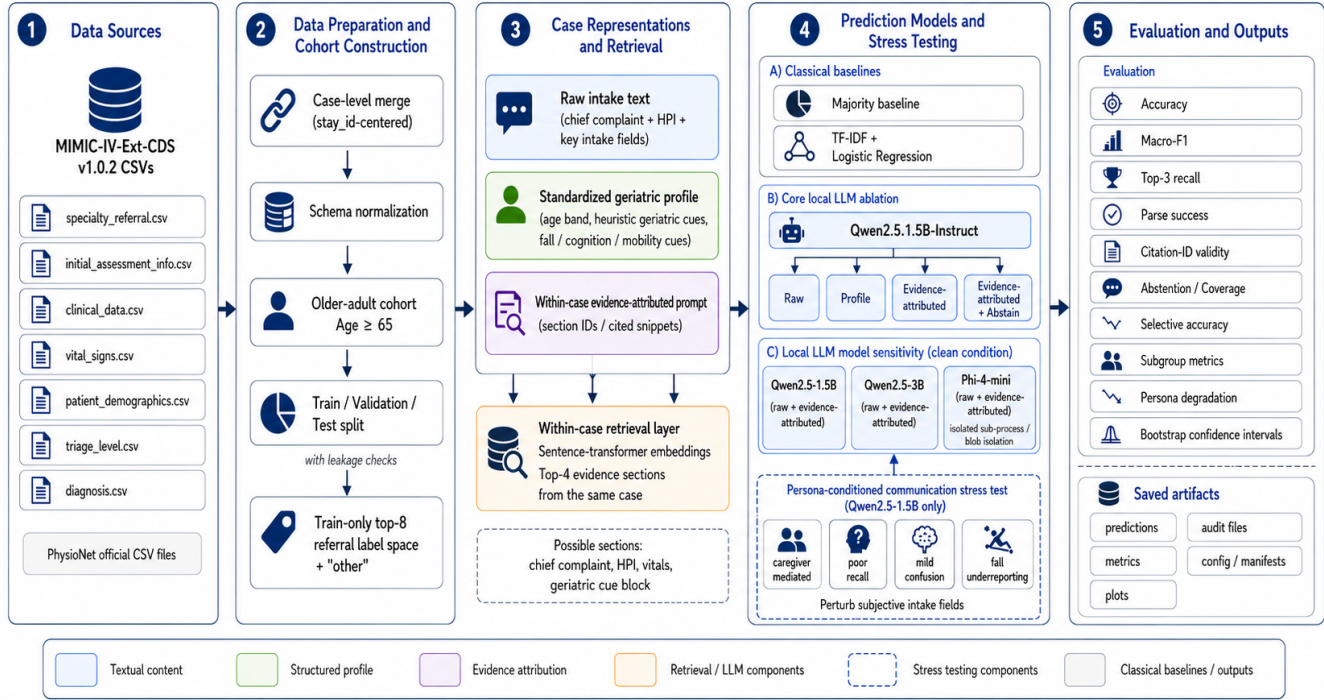


Figure 1: Core end-to-end architecture of the older-adult referral-support evaluation pipeline. The workflow constructs an older-adult cohort from MIMIC-IV-Ext-CDS, defines the referral label space using the training split only, evaluates classical baselines and local LLM prompting workflows, and stress-tests LLM behavior under deterministic communication perturbations. The local model-sensitivity extension is shown as a clean-condition sensitivity track and reported quantitatively in Table 3.

vitals, and a heuristic geriatric-cue section. This supports auditability by allowing a reviewer to check whether cited section IDs were present in the prompt. It does not establish that the evidence clinically supports the predicted specialty.

4.6 Baseline Models

We implemented two baselines. The majority baseline always predicts the most frequent training-set label. The TF-IDF logistic-regression baseline uses unigram and bigram TF-IDF features with class-balanced logistic regression. This baseline is essential because specialist referral labels may be highly associated with lexical cues in chief complaint and HPI. A clinically meaningful LLM workflow should be evaluated against such simple, reproducible baselines before claiming added value.

4.7 Local LLM Prompting Workflows

The core LLM workflow used Qwen2.5-1.5B-Instruct locally in Google Colab. The local setup avoids sending MIMIC-derived text to external LLM services, following PhysioNet responsible-use guidance [13]. Each prompt requested structured JSON with final referral label, ranked top-three labels, confidence, abstention flag, cited evidence IDs when applicable, and a brief rationale.

The full core prompt-arm comparison was performed for Qwen2.5-1.5B-Instruct:

- (1) raw prompting;
- (2) standardized-profile prompting;
- (3) within-case evidence-attributed prompting; and
- (4) evidence-attributed prompting with abstention enabled.

4.8 Local Model-Sensitivity Analysis

To test whether the clean-condition findings were specific to Qwen2.5-1.5B, we added a local model-sensitivity analysis. This compared Qwen2.5-1.5B-Instruct, Qwen2.5-3B-Instruct, and Phi-4-mini-instruct under two arms: raw prompting and within-case evidence-attributed prompting. The sensitivity analysis was limited to these two arms to keep runtime feasible in Colab.

Qwen2.5-1.5B predictions were reused from the core workflow. Qwen2.5-3B was loaded with 4-bit quantization for memory safety. Phi-4-mini was executed in an isolated subprocess with row-level checkpointing and dependency checks for the Transformers runtime, cache utilities, and loss-argument support. This isolation prevents a Phi-specific runtime failure from destroying already completed Qwen outputs. Phi-4-mini completed successfully in the reported run.

4.9 Communication Perturbation Stress Test

We created deterministic perturbations of the subjective history channel: poor recall, mild confusion, caregiver-mediated report, and

fall underreporting. These perturb chief complaint and HPI while preserving objective fields such as vitals. This is not a validated patient simulator. It is a reproducible stress-test layer for evaluating whether model behavior changes when older-adult communication becomes degraded.

The stress test was performed for Qwen2.5-1.5B profile and evidence-attributed arms. In the evidence-attributed stress-test arm, persona perturbations were applied to the standardized profile, while retrieval was intentionally performed from the original clean case sections. Therefore, this arm evaluates whether clean within-case evidence can buffer degraded subjective communication; it does not evaluate whether retrieval itself remains robust under fully degraded communication.

4.10 Evaluation Metrics

The primary metrics are accuracy and macro-F1. We also report top-3 recall, parse success, citation-ID validity, abstention rate, coverage, selective accuracy, subgroup metrics, and persona degradation. Bootstrap confidence intervals are reported for core clean-condition and model-sensitivity metrics where available.

Citation-ID validity measures whether cited evidence IDs were present in the retrieved evidence list. It does not measure clinical evidence support or rationale faithfulness. A manual audit file was exported for future review, but clinician-adjudicated evidence-support evaluation was not completed in this submission.

5 Experimental Design

5.1 Core Clean-Condition Comparison

The core experiment compared majority baseline, TF-IDF logistic regression, and four Qwen2.5-1.5B prompt arms on the same 121-case test set. The purpose was to test whether local LLM prompting improves over strong lexical baselines and whether standardized profiles or within-case evidence attribution improve clean referral prediction.

5.2 Local Model-Sensitivity Experiment

The model-sensitivity experiment tested whether the clean-condition findings were specific to Qwen2.5-1.5B. It compared Qwen2.5-1.5B, Qwen2.5-3B, and Phi-4-mini under raw and evidence-attributed prompting. This should be interpreted as a practical local-deployment sensitivity analysis rather than a perfectly controlled model-capacity experiment, because Qwen2.5-3B and Phi-4-mini required 4-bit quantization and Phi-4-mini was executed through a separate isolated process.

5.3 Communication Stress Experiment

The communication stress experiment evaluated deterministic older-adult communication perturbations. It was conducted only for the primary Qwen2.5-1.5B profile and evidence-attributed workflows. Therefore, persona-robustness findings should not be generalized to Qwen2.5-3B or Phi-4-mini.

Table 1: Cohort characteristics and field availability. The final analytic cohort contained 801 older-adult cases with no repeated subject IDs.

Category	Value
Source referral subset	2,200 cases
Age ≥ 65 before invalid-label removal	817 cases
Final older-adult analytic cohort	801 cases
Unique case IDs	801
Unique subject IDs	801
Train / validation / test	560 / 120 / 121
Mean age $\pm$ SD	76.70 $\pm$ 7.98
Median age	76
Age IQR	70–83
Age range	65–91
Age 65–74	365
Age 75–84	273
Age 85+	163
Female / male	428 / 373
Unique normalized referral labels	39
Chief complaint available	801 / 801
HPI available	801 / 801
Vitals available	778 / 801
Labs/tests available	796 / 801; excluded from main representation
PMH / medications available	0 / 801

6 Results

6.1 Cohort Characteristics

The final analytic cohort contained 801 older-adult cases. Table 1 summarizes cohort characteristics and input-field availability. Referral labels were imbalanced, with gastroenterology and orthopedics among the most frequent classes. Chief complaint and HPI were complete in the final cohort, while PMH, medications, assessment, and plan were unavailable in the extracted dataset. Vitals were available for most cases, and labs/tests were largely present but excluded from the main intake-stage representation. These characteristics motivate the use of macro-F1 and clarify the information boundary of the experiment.

6.2 Core Clean-Condition Performance

Table 2 shows clean-condition performance on the 121-case test set. TF-IDF logistic regression achieved the strongest performance, with accuracy 0.645 and macro-F1 0.588. The strongest Qwen2.5-1.5B arm was raw prompting, with accuracy 0.463 and macro-F1 0.346. Standardized profile prompting and within-case evidence-attributed prompting did not improve clean classification performance. The abstention-enabled arm had lower accuracy and macro-F1 than the other Qwen2.5-1.5B arms.

Figure 3 visually supports the primary empirical finding: TF-IDF logistic regression outperformed the core local Qwen2.5-1.5B prompting workflows under clean intake-stage conditions.

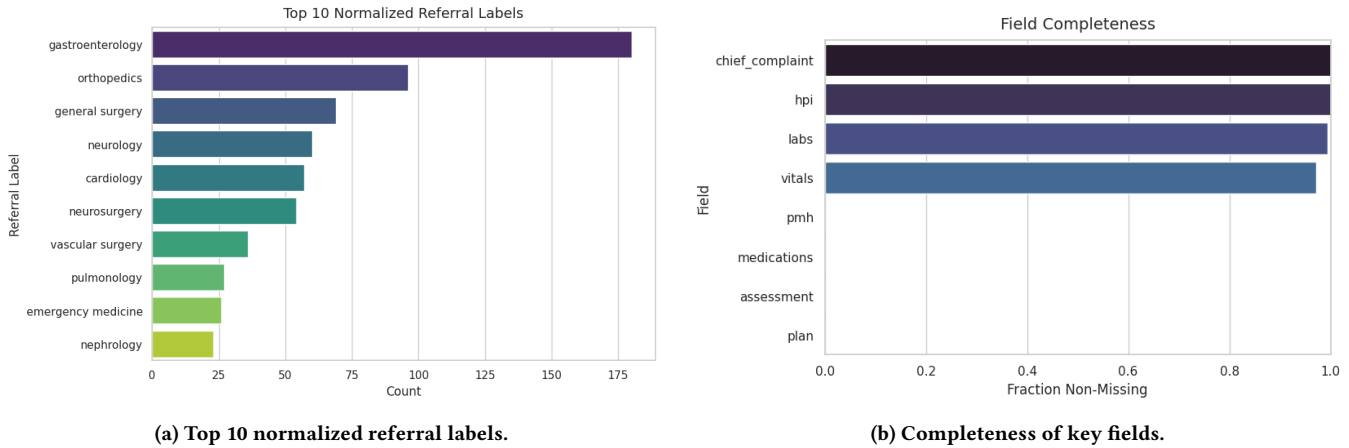


Figure 2: Cohort label distribution and input-field completeness. Referral labels were imbalanced. Chief complaint and HPI were complete, vitals were available for most cases, and labs/tests were largely present but excluded from the main intake-stage representation.

Table 2: Clean-condition performance on the 121-case test set. LLM top-3 recall is post-processed and exploratory because the parsed final label is forced into the first ranked position for consistency.

Model	<i>n</i>	Accuracy	95% CI	Macro-F1	95% CI	Top-3 Recall
Majority baseline	121	0.264	[0.182, 0.339]	0.046	[0.034, 0.057]	–
TF-IDF LR	121	0.645	[0.554, 0.736]	0.588	[0.490, 0.663]	0.884
Qwen2.5-1.5B Raw	121	0.463	[0.372, 0.554]	0.346	[0.257, 0.420]	0.669*
Qwen2.5-1.5B Profile	121	0.397	[0.314, 0.479]	0.257	[0.204, 0.305]	0.595*
Qwen2.5-1.5B Evidence-attributed	121	0.364	[0.281, 0.446]	0.279	[0.189, 0.353]	0.612*
Qwen2.5-1.5B Abstain	121	0.298	–	0.151	–	0.570*

*Post-processed exploratory top-3 recall. The evidence-attributed and abstention arms had citation-ID validity of 1.000. The abstention arm had coverage 0.893 and selective accuracy 0.324.

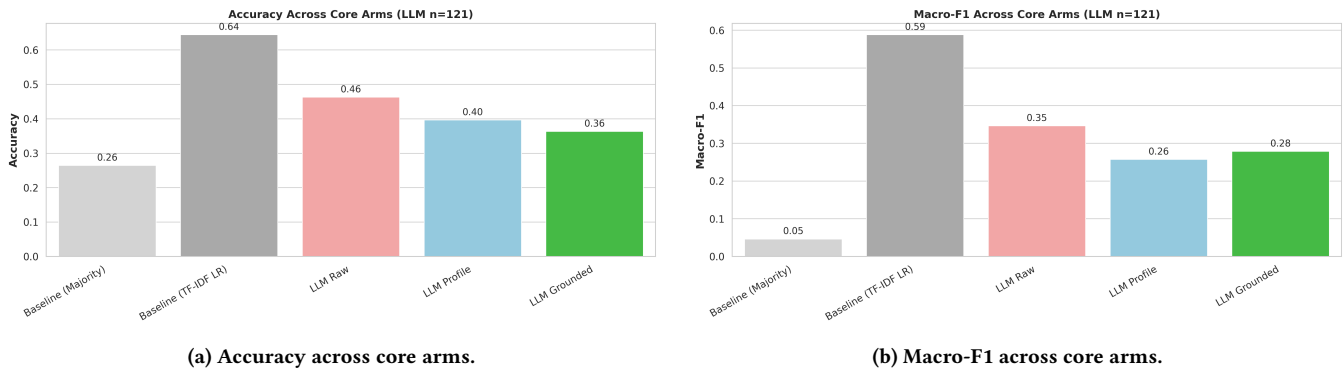


Figure 3: Clean-condition referral classification performance for majority, TF-IDF, and core Qwen2.5-1.5B arms. TF-IDF logistic regression achieved the strongest accuracy and macro-F1.

6.3 Local Model-Sensitivity Analysis

Table 3 reports the local model-sensitivity analysis. Scaling from Qwen2.5-1.5B to Qwen2.5-3B improved raw prompting from accuracy 0.463 to 0.537 and macro-F1 from 0.346 to 0.447. Phi-4-mini raw prompting achieved accuracy 0.496 and macro-F1 0.432. Despite

these improvements, neither Qwen2.5-3B nor Phi-4-mini exceeded the TF-IDF logistic-regression baseline.

Across all local LLM families, raw prompting outperformed evidence-attributed prompting on clean classification metrics. For Qwen2.5-1.5B, macro-F1 decreased from 0.346 to 0.279 under evidence attribution. For Qwen2.5-3B, macro-F1 decreased from 0.447

Table 3: Local model-sensitivity analysis. The sensitivity experiment evaluated raw and evidence-attributed prompting for Qwen2.5-1.5B, Qwen2.5-3B, and Phi-4-mini. TF-IDF LR is shown as the non-LLM reference baseline.

Model	Arm	<i>n</i>	Accuracy	95% CI	Macro-F1	95% CI	Top-3 Recall
TF-IDF LR	Baseline	121	0.645	[0.554, 0.736]	0.588	[0.490, 0.663]	0.884
Qwen2.5-1.5B	Raw	121	0.463	[0.364, 0.554]	0.346	[0.250, 0.428]	0.669*
Qwen2.5-1.5B	Evidence-attributed	121	0.364	[0.281, 0.446]	0.279	[0.191, 0.351]	0.612*
Qwen2.5-3B	Raw	121	0.537	[0.454, 0.620]	0.447	[0.344, 0.532]	0.818*
Qwen2.5-3B	Evidence-attributed	121	0.455	[0.364, 0.545]	0.368	[0.278, 0.442]	0.727*
Phi-4-mini	Raw	121	0.496	[0.413, 0.579]	0.432	[0.335, 0.516]	0.719*
Phi-4-mini	Evidence-attributed	121	0.488	[0.397, 0.579]	0.400	[0.317, 0.468]	0.686*

*Post-processed exploratory top-3 recall. Citation-ID validity was 1.000 for Qwen2.5-1.5B and Qwen2.5-3B evidence-attributed arms and 0.983 for the Phi-4-mini evidence-attributed arm.

to 0.368. For Phi-4-mini, macro-F1 decreased from 0.432 to 0.400. Thus, within-case evidence attribution improved traceability but did not improve clean predictive performance.

6.4 Evidence Attribution and Citation Validity

The evidence-attributed arms achieved high citation-ID validity. The Qwen2.5-1.5B and Qwen2.5-3B evidence-attributed arms achieved citation-ID validity of 1.000, and the Phi-4-mini evidence-attributed arm achieved 0.983. This means that cited evidence IDs were generally present in the retrieved evidence list. It does not prove that the cited evidence clinically supports the referral recommendation. A model can cite a valid section ID while still predicting the wrong specialty. Citation-ID validity is therefore an audit signal, not a measure of evidence faithfulness.

6.5 Abstention Behavior

The abstention-enabled Qwen2.5-1.5B arm had lower accuracy and macro-F1 than the other LLM arms. Its coverage was 0.893 and selective accuracy was 0.324. A useful abstention mechanism should improve accuracy among non-abstained cases by abstaining on uncertain or difficult cases. That did not occur. This indicates that simply adding abstention instructions to a prompt is insufficient for reliable selective prediction in this setting.

6.6 Communication Stress-Test Results

Table 4 shows performance under deterministic communication perturbations. Caregiver-mediated histories caused the largest degradation for both profile and evidence-attributed arms. In the profile arm, accuracy decreased from 0.397 to 0.256 and macro-F1 decreased from 0.257 to 0.167. In the evidence-attributed arm, accuracy decreased from 0.364 to 0.306 and macro-F1 decreased from 0.279 to 0.199. This was the largest macro-F1 degradation in both arms.

Figure 4 shows macro-F1 deltas across persona perturbations. Some perturbations produced small positive deltas. These should not be interpreted as degraded communication improving clinical reasoning. They may reflect prompt simplification, deterministic perturbation artifacts, or label-space effects. The most robust stress-test result is that caregiver-mediated history was consistently harmful.

7 Discussion

7.1 TF-IDF Remained the Strongest Clean-Condition Model

The central empirical finding is that TF-IDF logistic regression outperformed all local LLM conditions. This challenges the assumption that LLM prompting necessarily improves clinical NLP workflows. In this benchmark, lexical patterns in chief complaint and HPI appear highly informative. TF-IDF logistic regression exploited these cues more effectively than the local LLM prompting workflows.

This result is scientifically useful because it prevents overclaiming. Without a strong lexical baseline, the LLM arms might appear promising in isolation. With the baseline, the correct conclusion is more cautious: local LLM workflows produced structured outputs and rationales, but did not outperform classical text classification.

7.2 Model Sensitivity Strengthened the Negative Result

The model-sensitivity analysis reduces the risk that the result is specific to a single small model. Qwen2.5-3B and Phi-4-mini improved over Qwen2.5-1.5B, but neither closed the gap with TF-IDF logistic regression. This suggests that the referral task may be strongly lexical and that simple baselines remain essential in clinical NLP evaluation. The result should not be generalized to all larger medical LLMs, but it does show that modest local model scaling alone was insufficient in this setting.

7.3 Evidence Attribution Improved Audit Structure, Not Prediction

Within-case evidence attribution produced high citation-ID validity but did not improve clean classification performance. Across Qwen2.5-1.5B, Qwen2.5-3B, and Phi-4-mini, the raw prompt outperformed the evidence-attributed prompt on macro-F1. This does not mean evidence attribution is useless. It means its benefit should be framed as auditability and traceability rather than predictive performance.

This distinction is important for trustworthy clinical AI. Citation validity is not evidence faithfulness. A model can cite a valid section ID while still making an incorrect or weakly supported recommendation. Future work should add clinician review of whether cited evidence actually supports the predicted referral.

Table 4: Communication-stress performance and macro-F1 degradation for Qwen2.5-1.5B profile and evidence-attributed arms.

Arm	Perturbation	<i>n</i>	Accuracy	Macro-F1	Top-3 Recall*	Δ Accuracy	Δ Macro-F1
Profile	Caregiver-mediated	121	0.256	0.167	0.479	-0.140	-0.090
Profile	Fall underreport	121	0.405	0.278	0.595	+0.008	+0.021
Profile	Mild confusion	121	0.355	0.234	0.570	-0.041	-0.023
Profile	Poor recall	121	0.380	0.252	0.579	-0.017	-0.005
Evidence-attributed	Caregiver-mediated	121	0.306	0.199	0.388	-0.058	-0.080
Evidence-attributed	Fall underreport	121	0.339	0.266	0.554	-0.025	-0.013
Evidence-attributed	Mild confusion	121	0.339	0.224	0.554	-0.025	-0.055
Evidence-attributed	Poor recall	121	0.397	0.302	0.612	+0.033	+0.023

*Post-processed exploratory top-3 recall. Citation-ID validity for evidence-attributed stress arms was 1.000.

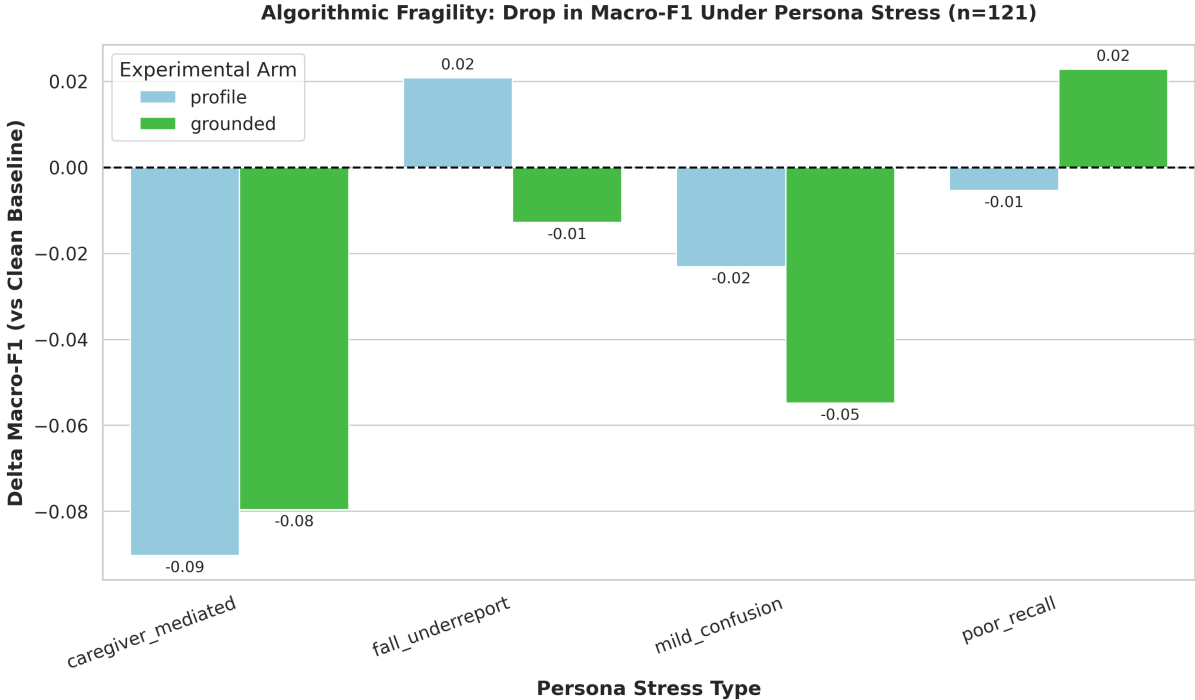


Figure 4: Macro-F1 degradation under older-adult communication perturbations. Caregiver-mediated histories caused the largest degradation for both standardized-profile and evidence-attributed workflows.

7.4 Caregiver-Mediated Histories Exposed Algorithmic Fragility

The caregiver-mediated perturbation was the most consistently harmful stress condition. This finding is clinically plausible. In older-adult care, histories may be indirect, incomplete, or filtered through caregivers. A referral-support model that relies heavily on subjective intake narratives may therefore become fragile when the communication channel changes.

The evidence-attributed arm provided only a limited and inconsistent buffer. For caregiver-mediated histories, evidence attribution reduced the accuracy drop compared with profile-only prompting, but macro-F1 still degraded substantially. Moreover, the grounded stress-test arm used clean within-case evidence as an intentional

buffer, so the result should not be interpreted as proof that evidence retrieval remains robust under degraded input.

8 Implications for Health Data Platforms

The caregiver-mediated failure mode has broader infrastructure implications. It suggests that older-adult referral-support systems should not rely only on a single narrative intake channel. Future health-data platforms could combine subjective intake text with longitudinal structured data, medication history, prior diagnoses, fall-event documentation, care-management notes, vital signs, laboratory results, and clinician-reviewed context. Such platforms may provide a more stable information substrate for AI-assisted referral support in aging-society settings.

This study does not prove that an integrated health-data platform will solve referral-support fragility. It motivates that research direction. The appropriate conclusion is that communication-stress failure modes provide a concrete reason to investigate longitudinal, structured, interoperable health-data infrastructure as part of future clinical AI systems.

9 Relation to Prior Work

Compared with MIMIC-IV-Ext-CDS and Gaber et al., this study is narrower but adds an older-adult robustness lens [4, 5]. Compared with medical QA benchmarks such as Med-PaLM and GPT-4 medical challenge evaluations, the task is closer to workflow-specific clinical text classification [12, 16]. Compared with RAG studies, the method is intentionally constrained: it retrieves within-case evidence rather than external knowledge [1, 8]. This makes the method privacy-preserving and auditable, but limits claims about medical grounding.

10 Limitations

10.1 Dataset Limitations

The cohort is derived from MIMIC-IV-Ext-CDS, not Japanese health-care data. Japan and other super-aging societies motivate the research framing, but the experiment does not validate the workflow in a Japanese clinical setting. Referral labels may not be clinician-adjudicated gold-standard labels for every case. The label space is collapsed into top-eight specialties plus *other*, simplifying the true referral problem. The main metrics use a primary label even though some cases may have multiple plausible referral specialties.

10.2 Model Limitations

The evaluated local models are small or moderate-sized instruction models selected for Colab feasibility. Results should not be generalized to all larger proprietary or medical-specialized LLMs. Qwen2.5-3B and Phi-4-mini were evaluated only in raw and evidence-attributed arms, not in the full profile, abstention, and persona-stress workflow. The standardized geriatric cues are heuristic and limited by missing PMH and medication fields.

10.3 Evaluation Limitations

LLM top-3 recall is post-processed and exploratory because the parser forces the final parsed label into the first ranked position. Therefore, accuracy and macro-F1 are the primary metrics. Citation-ID validity does not prove clinical evidence support or rationale faithfulness. Manual audit files were exported, but clinician-adjudicated evidence-support review was not completed. Persona perturbations are deterministic English-language transformations, not validated simulations of real patient or caregiver communication.

10.4 Stress-Test Limitation

In the evidence-attributed persona stress-test arm, retrieval used the original clean case sections while the profile text was persona-perturbed. This design intentionally tests whether clean within-case evidence can buffer degraded subjective communication. It does not test whether retrieval itself remains robust when all available text

is degraded. Future work should compare clean-evidence buffering, objective-only retrieval, and fully persona-perturbed retrieval.

10.5 Clinical Responsibility and Non-Deployment Statement

This study should not be interpreted as a clinical decision system or as evidence that a local LLM is ready for specialist referral support. The evaluated workflows are retrospective, notebook-based, and intended for audit and education. Any real clinical application would require prospective validation, clinician review, calibration, bias and subgroup analysis, privacy review, integration testing, and clear human accountability.

11 Future Work

Future work should evaluate larger local and medically tuned models, compare intake-stage and full-chart settings, test clinician-approved referral subsets, incorporate external guideline retrieval, improve abstention calibration, and conduct clinician adjudication of evidence support. A particularly important next experiment is to rerun the evidence-attributed stress test with persona-perturbed retrieval sections or objective-only retrieval, so that robustness claims are not dependent on clean subjective evidence being available.

Longer-term work should examine whether health-data platforms can reduce dependence on incomplete subjective histories by combining longitudinal structured data, prior diagnoses, medication history, fall history, care-management documentation, and objective measurements for aging-society care. This should be evaluated empirically rather than assumed from the current stress-test results.

12 Conclusion

This study developed a reproducible local-LLM evaluation pipeline for older-adult intake-stage specialist referral support using MIMIC-IV-Ext-CDS. The central empirical result is that TF-IDF logistic regression outperformed all local LLM conditions. Scaling from Qwen2.5-1.5B to Qwen2.5-3B and testing Phi-4-mini improved local LLM performance but did not close the gap with TF-IDF. Evidence attribution improved citation traceability but not clean classification. The abstention prompt did not create a reliable selective-prediction mechanism. Communication stress testing showed that caregiver-mediated histories were the most consistently harmful perturbation.

The study provides a responsible evaluation framework rather than an overclaimed clinical agent. It shows that local LLM health-care workflows should be benchmarked against simple baselines, evaluated under explicit information boundaries, stress-tested under older-adult communication degradation, and audited separately for evidence citation and clinical correctness. The health-data platform implication is best treated as a future research direction: caregiver-mediated degradation motivates investigation of longitudinal, structured, interoperable information substrates, but the present experiment does not prove that such platforms will solve referral-support fragility.

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